

* Important Algos for Placements : →

- * Kadane's Algo ✓ Maximum Subarray Sum
- * Sieve Of Eratosthenes → Prime numbers (1 to N)
- * Euclid's Division Algo → GCD using Recursion
- * Rabin Karp Algo → String Algo
- * Binary Search Applications
- * Recursion Applications ↔ DP Backtracking

DSA :

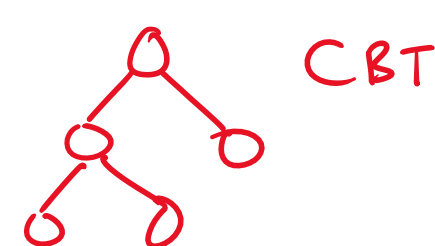
Algos : →

Searching

Linear $O(n)$
 Binary $O(\log(n))$
 Recursive Binary $O(\log(n))$
 Jump Search $O(\sqrt{n})$
 Interpolation Search → (pos)

Sorting

* Bubble $O(n^2)$
 * Selection (swap) $O(n^2)$
 * Insertion (shift) $O(n^2)$
 * Merge $n \log n$
 * Quick $n \log n$
 * Heap (CBT) $n \log n$
 * (ATML) Shell Sort
 NCA : { ① Radix Sort
 { ② Count Sort
 (Bucket Sort)



* Introduction to Object Oriented Programming

* OOPs was introduced to represent real-world entities.

Blueprint / Template → characteristics
 ↓
 class Student {
 String name;
 int id;
 int marks;
 }
 { Student → entity
 { name, id, marks → attributes or properties or instance variables
 { evaluateResult() { (method) (Behaviour)

Class: It is a blueprint / template to create objects. It has no memory of its own. It is just like a rule to define real world entities like — Student, Teacher, Employee, Cat.

Object: "An object is an instance of a class"

It is actually the key to access all the data inside a class.

An object has memory of its own & is created by using the "new" keyword in Java.

It occupies the heap memory. More objects — greater heap — less efficient application. It is recommended to not create unnecessary objects.

(Reference = Object = Instance) (Same)

Method: Functions in C & C++ are called methods in Java, because they describe the behaviour of the objects of the class.

DRY: Don't Repeat Yourself

Constructor: It is a special method used to initialize or instantiate objects of a class.

It is of two types:

- a) default Constructor / (No-argument Constructor)
 → When we don't create our constructor, the compiler (JVM) creates its own default constructor.
- b) Parameterized Constructor:
 When the user defines the constructor, it is the parameterized constructor.
 * When parameterized constructor is created, the default constructor gets deleted ****

Rules for a constructor : →

1. A constructor is always public or default.
2. It is the same name as the class followed by a parentheses — Class Name ();
3. It has no return type.
4. There can be multiple constructors inside a class. (Constructor Overloading)

Rules of Constructor Chaining : (Interviews)

- this() should be the first statement in the constructor.
- Used to call another constructor of same class.
- Prevent "code duplication".
- Improves codebase maintainability.
- Common in real-world frameworks like (Spring, Hibernate, Spring Boot etc)

"Constructor chaining allows one constructor to call another constructor using this(), ensuring centralized initialization and eliminating code duplication."